TEXMART – SMART TEXTILE MANAGEMENT SYSTEM

A MINI PROJECT REPORT

22ISP61 – MINI PROJECT

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in partial fulfilment of the
requirements for the award of
the degree of

MASTER OF SCIENCE

IN

SOFTWARE SYSTEMS

DEPARTMENT OF COMPUTER TECHNOLOGY – PG

KONGU ENGINEERING COLLEGE

TRANSFORM YOURSELF

(Autonomous)

PERUNDURAI ERODE – 638060

APRIL 2026

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APRIL 2026

BONAFIDE CERTIFICATE

This is to certify that the mini project report entitled "**TEXMART - SMART TEXTILE MANAGEMENT SYSTEM**" is the bonafide record of project work done by **PRANITHA SENTHILMURUGAN GEETHA (Register No. 23ISR044), SHALINI S R (Register No. 23ISR053), VARSHASRI K (Register No. 23ISR060)** in partial fulfilment of the requirements for the award of the Degree of Master of Science in Software Systems of Anna university, Chennai during the year 2025 – 2026.

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DATE:

Submitted for the end semester viva voce examination held on_____

INTERNAL EXAMINER **EXTERNAL EXAMINAR**

DECLARATION

I affirm that the mini project report titled "**TEXMART – SMART TEXTILE MANAGEMENT SYSTEM"** being submitted in partial fulfilment of the requirements for the award of Master of Science is the original work carried out by me. It has not formed the part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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I certify that the declaration made by the above candidate is true to the best of my knowledge.

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ABSTRACT

The TexMart – Smart Textile Management System is designed to streamline and modernize the management of textile businesses by addressing the limitations of traditional manual and semi-digital processes. Textile businesses often face challenges in maintaining inventory records, tracking sales, managing suppliers, and analyzing business performance efficiently. These issues can lead to errors, delays, and reduced productivity, ultimately affecting overall business growth and customer satisfaction. The proposed system provides a comprehensive and user-friendly digital platform that enables efficient management of textile operations. It offers features such as real-time inventory tracking, automated billing, supplier management, and detailed sales analytics. The system allows users to easily update stock, monitor product availability, and generate reports for better decision-making. Additionally, it enhances operational efficiency by reducing manual workload and minimizing errors through automation. TexMart is developed using modern web technologies, including HTML, CSS, JavaScript, and React for the front end, along with Node.js and Express.js for the back end. MongoDB is used for efficient data storage and retrieval. The system ensures data security, scalability, and high performance, making it suitable for both small-scale and large-scale textile businesses. Overall, TexMart provides an effective solution to improve productivity, accuracy, and business insights in the textile industry.

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ORGANIZATION PROFILE

Founded with the vision of transforming traditional textile business operations through digital innovation, TexMart – Smart Textile Management System represents a modern approach to managing textile enterprises efficiently and intelligently. The system is designed to bridge the gap between conventional management practices and the growing demand for technology-driven solutions in the textile industry.

TexMart has established itself as a reliable platform by providing a comprehensive solution for handling key business activities such as inventory management, sales tracking, billing, and supplier coordination. A distinctive aspect of TexMart is its focus on automation and real-time data management. The system empowers users with features such as automated stock updates, detailed reporting, and insightful analytics that support informed decision-making. By simplifying complex processes and reducing manual effort, TexMart enhances productivity and operational efficiency.

In addition to its core functionalities, TexMart emphasizes user-friendly design and accessibility, ensuring that business owners and staff can easily adapt to the system without extensive technical expertise. Built on modern web technologies, it offers scalability, security, and high performance to support businesses of varying sizes. With a strong foundation in innovation, reliability, and efficiency, TexMart continues to serve as a smart solution for textile management. It stands as a testament to how digital transformation can redefine traditional business practices, enabling growth, accuracy, and improved customer satisfaction in the textile sector.

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To
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Dear Sir/Madam,

Sub: Confirmation - Consultancy Work - Regarding

With reference to the discussions we had with Dr. Maragatham T., Associate Professor, Computer Technology PG, we are pleased to offer the consultancy work for software development to your college. We acknowledge that the work will be executed by Pranitha S G. (Reg No: 231SR044), Shalini S R. (Reg No: 231SR053), and Varshasri K. (Reg No: 231SR060) from III year M.Sc. Software Systems. This arrangement is effective from December 2025 and will remain valid until March 2026.

Thank You

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

The **"TEXMART – SMART TEXTILE MANAGEMENT SYSTEM"** is a user-friendly web-based platform designed to streamline and digitize textile business operations. It integrates inventory management, sales tracking, billing, and supplier coordination into a single system, enabling businesses to efficiently manage their day-to-day activities with improved accuracy and reduced manual effort.

A key feature of the system is its Real-Time Inventory Management, which allows users to monitor stock levels, update product details, and track availability instantly. This is complemented by an Automated Billing System, which simplifies transactions, generates invoices, and ensures accurate financial records. Additionally, the system provides Sales Analytics and Reporting, offering valuable insights into product performance, revenue trends, and customer purchasing patterns.

The platform also includes a Supplier Management Module, enabling businesses to maintain supplier details, track orders, and manage procurement efficiently. On the administrative side, the Admin Panel provides complete control over products, inventory, sales records, suppliers, and user data. It also supports the generation and download of detailed reports, helping administrators analyse business performance and make informed decisions.

Overall, TexMart creates a smart and efficient digital environment where textile business operations are simplified through technology, enhancing productivity, accuracy, and overall business growth.

CHAPTER 2

SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

Traditional textile businesses often depend on manual methods or basic digital tools to manage inventory, sales, and supplier details. These systems provide only simple record-keeping without real-time updates or advanced features. As a result, managing daily operations becomes time-consuming and less efficient. Inventory tracking is often inaccurate, leading to problems like excess stock or shortages. Billing and sales processes are handled manually, increasing the risk of calculation errors. There is also no proper system to analyse sales data or business performance. Information is usually scattered, making it difficult to access and manage. These limitations reduce productivity and slow down decision-making. Businesses struggle to adapt to modern demands due to lack of automation. Overall, such systems are outdated and not suitable for efficient textile management.

Drawbacks of Existing System

- Lack of Automation
- Inefficient Inventory Management
- Manual Billing Errors
- Limited Data Analysis

2.2 PROPOSED SYSTEM

The TexMart – Smart Textile Management System offers real-time inventory management that allows users to track and update stock levels easily. Its automated billing system enhances efficiency by generating accurate invoices and reducing manual errors. The platform simplifies business operations by integrating inventory, sales, and supplier management into a single system. Additionally, it provides detailed reports and analytics, helping business owners make better decisions. The user-friendly interface ensures easy navigation and improves overall productivity

Advantages of Proposed System

- Improves inventory management
- Enhances data accuracy
- Reduces manual workload
- Increases efficiency with automation and real-time tracking

2.3 FEASIBILITY STUDY

The feasibility study involves a detailed analysis carried out during the development of the TexMart – Smart Textile Management System. Each component of the system is carefully evaluated to ensure it meets the needs of textile business owners in an efficient and user-friendly manner. It is important to identify the type of data required, as the system focuses on managing inventory, sales, and supplier information effectively. System analysis involves collecting, organizing, and evaluating data related to business operations and workflow. This ensures that the proposed system is practical, reliable, and capable of improving overall performance.

Economical Feasibility

The economic feasibility of implementing the TexMart – Smart Textile Management System involves evaluating the initial investment in system development, software, and basic hardware requirements, along with ongoing costs for maintenance and support. While the initial cost may include setting up features such as inventory management, automated billing, and reporting tools, the long-term financial benefits outweigh these expenses. By digitizing business operations, the system reduces manual work, paperwork, and the chances of costly errors. Additionally, improved inventory control and faster billing processes help in reducing losses and increasing operational efficiency. The system also provides insights into sales and stock trends, enabling better decision-making and resource utilization. Overall, TexMart contributes to cost savings, improved productivity, and increased profitability for textile businesses.

Operational Feasibility

Operational feasibility is a key factor in the successful implementation of the TexMart – Smart Textile Management System. Designed with simplicity and ease of use in mind, the system ensures a smooth experience for textile business owners and staff. It simplifies tasks such as inventory management, billing, and supplier tracking, reducing dependence on manual

processes. The automated features help minimize workload and improve efficiency in daily operations.

Integration with existing workflows allows businesses to adopt the system without major disruptions. The system can handle large volumes of product data and transactions, making it suitable for growing textile businesses. Additionally, it is scalable and flexible, allowing easy updates and future enhancements. Proper training and support enable users to operate the system effectively, improving overall productivity and business performance.

Technical Feasibility

The technical feasibility of the TexMart – Smart Textile Management System is based on its compatibility with modern web technologies and its ability to support future enhancements. The system is developed using reliable frameworks that ensure high performance, smooth operation, and secure data handling. It supports responsive design, making it accessible across devices such as desktops, tablets, and mobile phones for convenient usage. The system uses a robust database to efficiently manage inventory, sales records, and supplier information. Its architecture supports real-time data processing, ensuring quick updates and accurate information. Integration of features like automated billing and reporting requires a stable and scalable technical infrastructure. Regular updates, security measures, and technical support ensure the system remains reliable and up-to-date, making it a sustainable solution for textile business management.

2.4 PROBLEM ANALYSIS

In the traditional approach to managing textile business operations, several challenges arise due to manual and fragmented processes. These include the risk of inaccurate or outdated inventory records, which can lead to confusion in stock management and order handling. Additionally, the absence of an integrated digital system often results in delays and errors in billing, sales tracking, and supplier coordination, negatively impacting overall efficiency. Without a centralized platform, business owners have limited visibility into stock levels and sales performance, making decision-making difficult. This lack of automation and real-time access to information reduces productivity and can hinder business growth in a competitive textile market.

2.5 DATA FLOW DIAGRAM

The Data Flow Diagram (DFD) of the TexMart – Smart Textile Management System illustrates the flow of data between users, administrators, and system processes. It highlights key functions such as inventory management, billing, sales tracking, and supplier management, showing how data moves within the system and interacts with different components.

Context Diagram

The context diagram for the TexMart – Smart Textile Management System represents a centralized system that interacts with users, suppliers, and administrators. Users can manage inventory, generate bills, and track sales efficiently through the system. Suppliers contribute by providing product details and managing supply information. Administrators oversee product data, inventory levels, sales records, and supplier details. This overview reflects the system's digital approach, focusing on streamlined operations, real-time data management, and reducing dependency on manual processes.

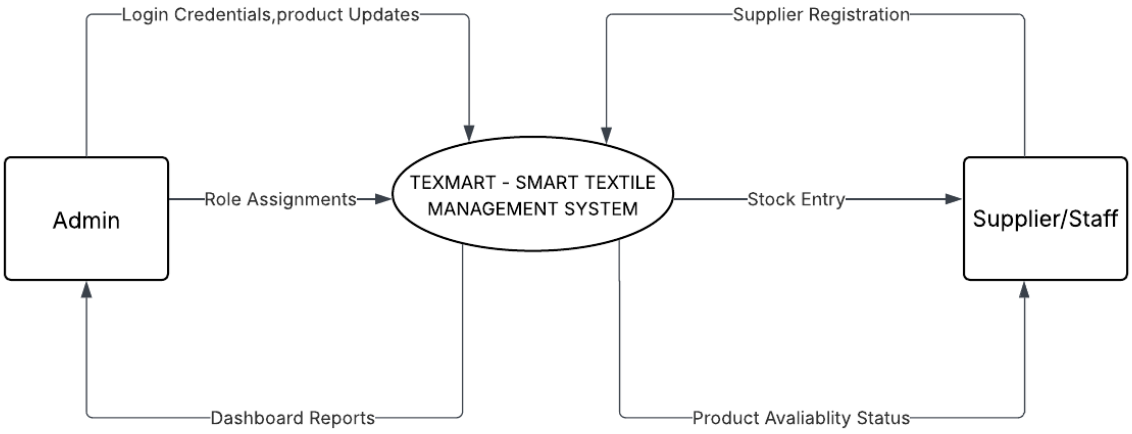


Figure 2.1 Level 0 DFD -Texmark Smart Textile Management System

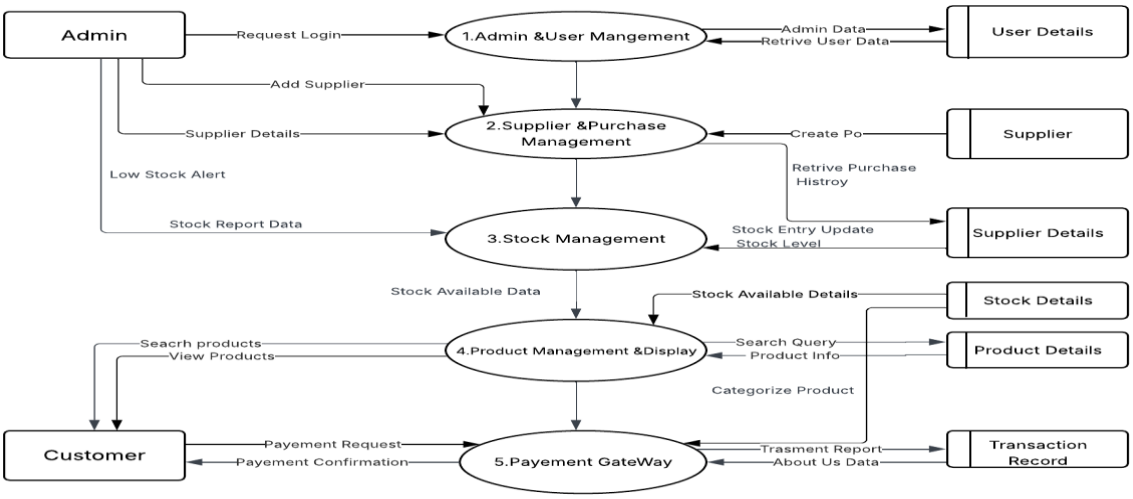


Figure 2.2 Level 1 DFD -Texmark Smart Textile Management System

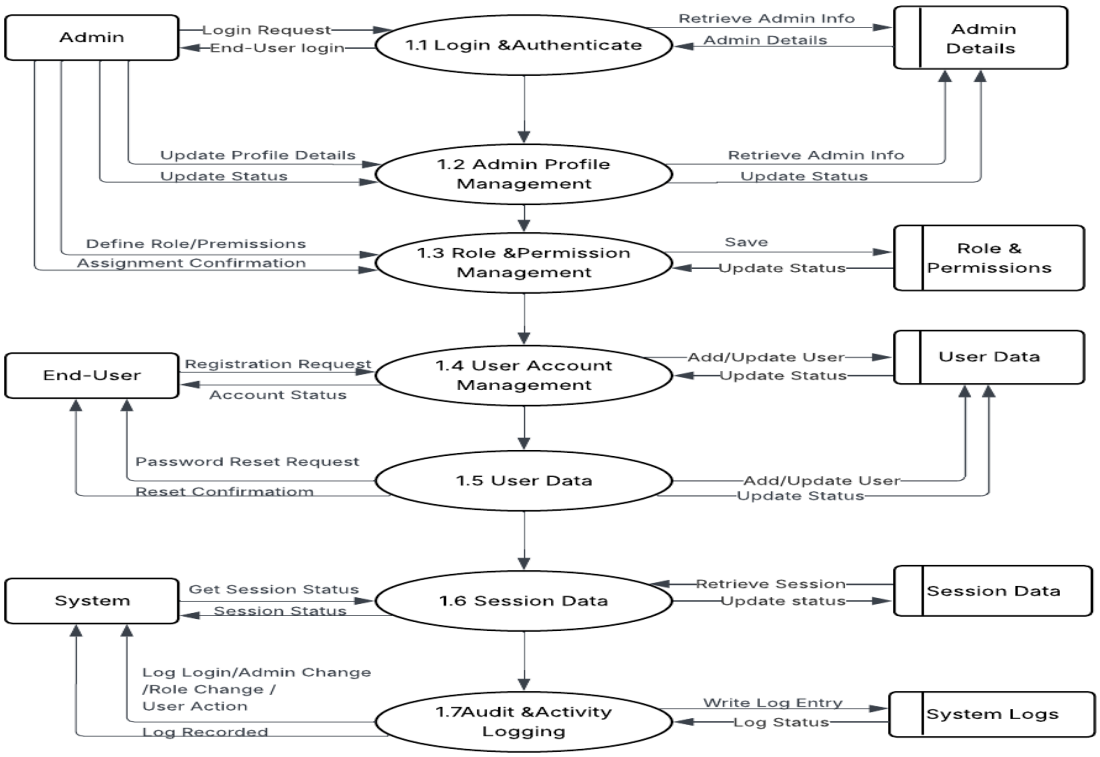


Figure 2.3 Level 2 DFD -Admin &User Management

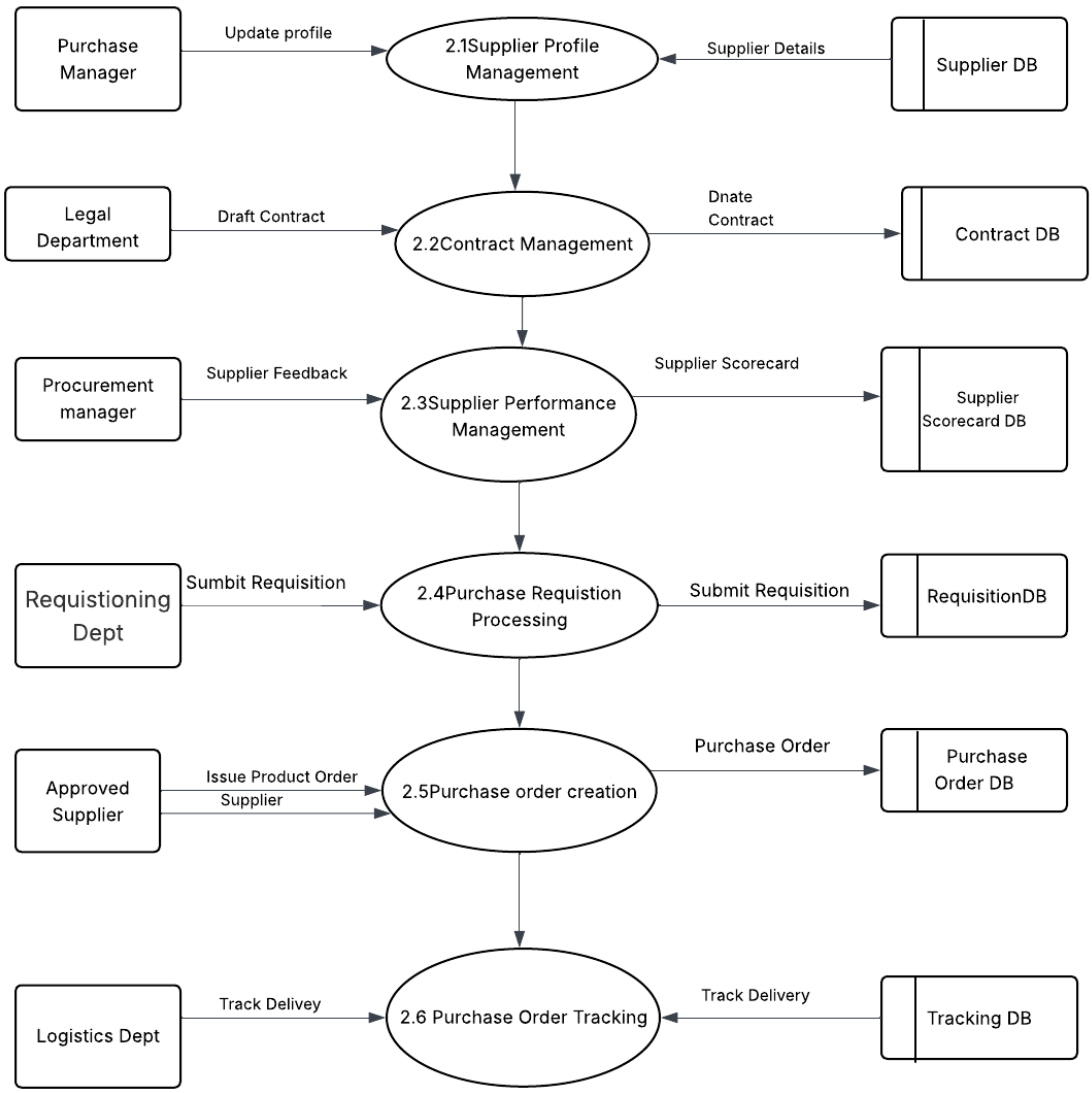


Figure 2.4 Level 2 DFD(Supplier and Purchase Module)

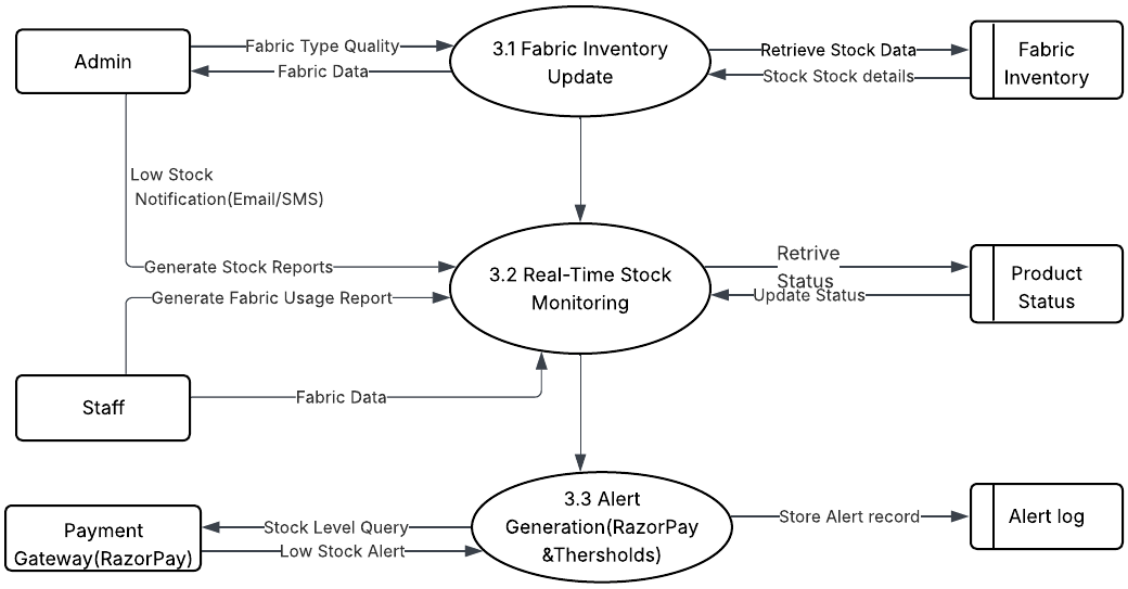


Figure 2.5 Level 2 DFD(Stock Management Module)

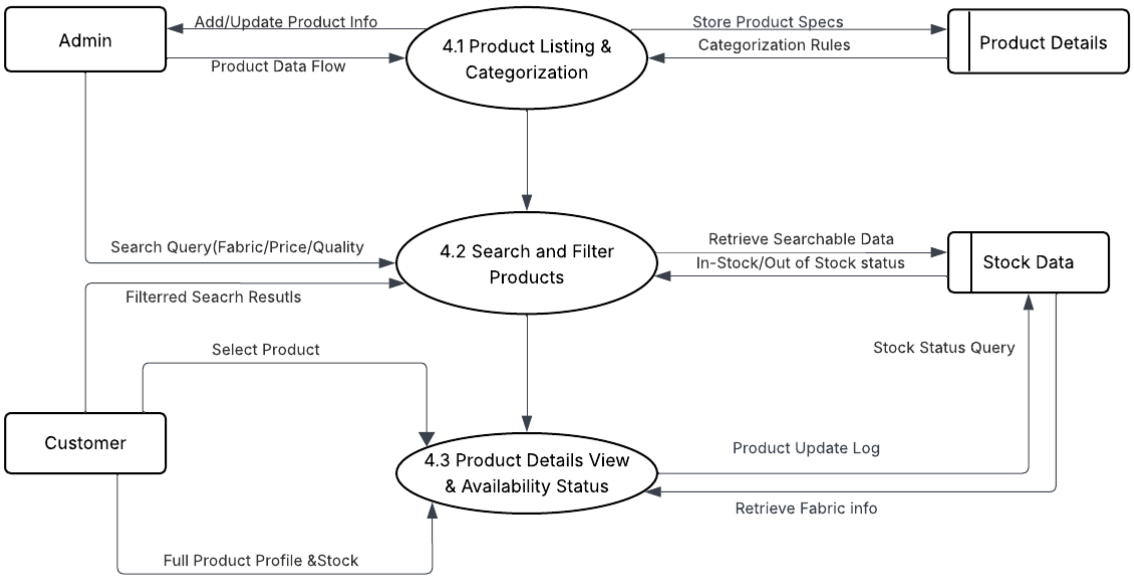


Figure 2.6 Level 2 DFD(Product Display Module)

2.6 SYSTEM SPECIFICATION

Hardware Requirements

This section gives the details and specification of the hardware on which the system is expected to work.

Processor	:	Intel Core i3 / above
Hard Disk	:	500 GB
RAM	:	4 GB DDR4 or above
Monitor	:	15” Color Monitor
System Type	:	64-bit Operating System

Software Requirements

This section gives the details and specification of the software on which the system is expected to work.

Front-End	:	HTML, CSS, Bootstrap 5
Client-side Script	:	JavaScript
Back-End	:	MongoDB
Database	:	MySQL
Operating System	:	Windows 11

2.7 SOFTWARE DESCRIPTION

HTML

HTML stands for Hyper Text Markup Language, which is used to describe the structure of web pages. HTML elements act as the building blocks of a webpage. It is a markup language

that web browsers use to display text, images, buttons, forms, tables, and other content in a visual format.

In this project, HTML is used to design the main pages such as Login Page, Dashboard, Project Management Page, Module Management Page, Ticket Creation Page, Assignment Page, and Status Tracking Page. It helps in creating a proper structure for user interaction and system navigation.

Features of HTML include the following:

- It is easy to understand and modify.
- It helps in creating effective web page layouts.
- It supports forms, tables, and hyperlinks.
- It can be displayed on all platforms such as Windows and Linux.
- Images, icons, and multimedia can be added easily.

CSS(Cascading Style Sheet)

CSS stands for Cascading Style Sheet. It is a style sheet language used to describe the presentation and visual appearance of web pages created using HTML. CSS helps in separating the content structure from the design and layout, which makes the web pages easier to manage and maintain.

In this project, CSS is mainly used to improve the appearance of the dashboard, ticket creation forms, project list pages, module management pages, assignment pages, status tracking pages, and report pages. It helps in designing the layout of buttons, tables, menus, headings, input fields, and navigation panels in a clear and organized manner.

CSS plays an important role in making the application user-friendly and visually attractive. It is used to define colors, fonts, text styles, margins, padding, alignment, borders, and spacing between elements. By using CSS, the system interface becomes neat, consistent, and easy for users to understand.

Another major advantage of CSS is that it supports responsive web design, which means the pages can adjust properly to different screen sizes such as desktop, laptop, tablet, and mobile devices. This improves the accessibility of the system.

In the Task Manager Ticketing Tool, CSS also helps in highlighting important sections such as priority labels, status indicators, alerts, notifications, and dashboard cards, making it easier for users to track tasks and project progress.

Features of CSS include the following:

- Controls font style, size, and colors
- Improves page layout and alignment
- Makes pages responsive and device-friendly
- Supports reusable design styles
- Enhances overall user experience
- Provides consistent design throughout the system

MongoDB

MongoDB is a NoSQL database known for its flexibility, scalability, and document-based storage. It stores data in JSON-like documents, allowing dynamic data structures without requiring a fixed schema. This makes it well-suited for the TexMart – Smart Textile Management System, where data such as inventory details, sales records, and supplier information may vary in structure.

Key benefits include:

• Schema-less design for easy updates and flexible data handling

• Horizontal scalability to manage growing business data

• Fast queries and indexing for real-time data access

• High availability through replication and data backup support

MongoDB supports TexMart's requirement for efficient data storage, quick updates, and smooth scalability, ensuring reliable performance for textile business operations.

MySQL

MySQL is an open-source relational database management system (RDBMS) that is used to store, organize, and manage data efficiently. It uses Structured Query Language (SQL) for performing operations such as inserting, updating, deleting, and retrieving data from the

database. MySQL is widely used in web application development because it is fast, reliable, secure, and capable of handling large amounts of data.

In the Task Manager Ticketing Tool for Project and Team Management, MySQL is used as the backend database to store all the important information such as user details, project details, module details, ticket information, assignment records, status tracking data, and dashboard reports. It works efficiently with PHP and supports all CRUD operations required for the project. MySQL helps in maintaining data consistency, quick retrieval of records, and secure storage of project-related information, making it highly suitable for this web-based management system.

CHAPTER 3

SYSTEM DESIGN

3.1 INPUT DESIGN

Input design is the process of converting user-provided data into a format that the system can easily understand and process. It plays a vital role in the effectiveness of the TexMart – Smart Textile Management System and is one of the most critical phases in system development. Poor input design can lead to incorrect data entry, processing errors, and reduced efficiency in business operations.

In TexMart, input design ensures that users and administrators can interact with the system smoothly and efficiently. It focuses on creating a user-friendly interface for entering data such as product details, inventory updates, sales transactions, and supplier information. The goal is to ensure that data entry is accurate, validated, and easy to perform. The system includes the following input constraints:

• All required data such as inventory, sales, and supplier details must be accurately captured and processed.

• Forms and menus should be well-structured, simple to navigate, and properly formatted.

• Inputs should be validated to prevent errors such as incorrect entries or missing information.

• The design should ensure data clarity, availability, and reliability across the system.

Effective input design improves data accuracy, reduces errors, and ensures smooth functioning of the TexMart system.

Input Forms Used in the System

Adding user details

The system includes simple forms to collect essential business and user information. Users are required to enter details such as product name, quantity, price, supplier information, and billing data. These inputs are stored in the system to manage inventory, generate invoices, and track sales activities. This helps streamline operations and ensures accurate record-keeping within the TexMart system.

Adding Textile Products

The system allows admins to add textile products by entering details such as product name, category, quantity, price, and supplier information. These products are then stored in the system and displayed in the inventory section for easy management and tracking. This helps in maintaining accurate stock records and ensures smooth textile business operations.

Module Details Form

This form is used to create and manage modules under each project. It contains fields such as module name, module description, assigned team member, start date, due date, and module status. These inputs help in dividing the project into manageable parts and tracking progress easily.

3.2 DATABASE DESIGN

The most important consideration in designing the database is how information will be stored and used efficiently. The main objective of designing the database is to store project-related data in an organized manner and retrieve it quickly whenever required.

The database for this project includes tables such as:

- Admin Details
- Customer Details
- Product Details
- Category Details
- Product Purchase Details
- Payment Details

- Customer Feedback

The database is designed using MySQL to ensure data consistency, easy retrieval, and efficient management of project and task records.

Table 3.1 Admin Details

FIELD NAME	TYPE	SIZE	CONSTRAINTS
Admin_id	Integer	4	Primary Key
Password	Varchar	10	Not Null

Table 3.2 Customer Details

FIELD NAME	TYPE	SIZE	CONSTRAINTS
Customer_id	Integer	4	Primary Key
Customer_Name	Varchar	10	Not Null
Customer_Password	Varchar	12	Not Null
Customer_Email	Varchar	30	Not Null

Table 3.3 Product Details

FIELD NAME	TYPE	SIZE	CONSTRAINTS
Product_id	Integer	4	Primary Key
Product_Name	Varchar	10	Not Null

Product_Description	Varchar	60	Not Null
Product_Price	Double	8	Not Null
Product_Image	Varchar	50	Not Null
Category_Id	Varchar	15	Foreign Key
Product_Brand	Varchar	15	Not Null

Table 3.4 Category Details

FIELD NAME	TYPE	SIZE	CONSTRAINTS
Category_id	Integer	4	Primary Key
Category_Name	Varchar	10	Not Null

Table 3.5 Product Purchase Details

FIELD NAME	TYPE	SIZE	CONSTRAINTS
Purchase_id	Integer	4	Primary Key
Customer_id	Integer	4	Foreign Key
Product_id	Integer	4	Foreign Key
Purchase_Date	DateTime		Not Null
Shipping_Address	Varchar	20	Not Null

Amount	Decimal	8	Not Null
Receipt_Number	Varchar	50	Unique

Table 3.6 Payment Details

FIELD NAME	TYPE	SIZE	CONSTRAINTS
Payment_id	Integer	4	Primary Key
Customer_id	Integer	4	Foreign Key
Purchase_id	Integer	4	Foreign Key
Purchase_Date	DateTime		Not Null
Amount	Decimal	8	Not Null
Receipt_Number	Varchar	50	Unique

Table 3.7 Customer Feedback

FIELD NAME	TYPE	SIZE	CONSTRAINTS
Customer_id	Integer	4	Foreign Key
Product_id	Varchar	20	Not Null
Feedback_Description	Varchar	255	Not Null
Feedback_Rating	Integer	4	Not Null

3.3 MODULES DESCRIPTION

The main modules in this project are listed below

- Admin Panel Module
- Supplier and Purchase Module
- Stock Management Module
- Product Display Module

Admin Panel Module

The Admin Panel serves as the control hub for the TexMart – Smart Textile Management System, enabling administrators to manage all system activities efficiently. It provides a user-friendly interface with dashboard and reports to monitor overall operations and performance. Admins can handle user login and authentication, manage roles, and control access within the system.

The panel allows administrators to add, update, and delete product details, ensuring accurate inventory management. It also supports managing supplier information, stock updates, and system records. With integrated reporting features, admins can track data, generate reports, and analyse business performance. Overall, the Admin Panel ensures smooth system operation, improved control, and efficient management of textile business processes.Sub Modules:

- User Registration
- User Login
- Profile Management
- Notification Management

Supplier and Purchase Module

This module serves as a core component of the TexMart – Smart Textile Management System, focusing on efficient handling of supplier and purchasing activities. It provides a structured environment where administrators can register and manage supplier details, ensuring smooth coordination and communication. The module enables the creation of purchase orders, allowing businesses to procure materials and products in an organized manner.

It also maintains purchase history records, helping track past transactions and supplier performance. Additionally, the module manages invoice and payment records, ensuring

accurate financial tracking and transparency. With a user-friendly interface and integrated features, it simplifies procurement processes and improves efficiency. Acting as a bridge between suppliers and the business, this module ensures timely procurement, better resource management, and streamlined operations.

Stock Management Module

Designed for efficient inventory control and monitoring, this module plays a vital role in managing stock within the TexMart – Smart Textile Management System. It provides a structured environment for handling stock entry, updates, and tracking of textile products in real time. Users can easily monitor stock levels, ensuring accurate availability of materials and products at all times.

The module includes features such as low stock alerts, helping administrators take timely actions to avoid shortages. It also maintains stock reports and history, enabling better analysis of inventory usage and trends. Admins can manage and update stock details seamlessly through a user-friendly interface. This module enhances accuracy, reduces manual errors, and ensures smooth inventory management, making it essential for efficient textile business operations.

Product Display Module

The Product Display Module serves as a key component of the TexMart – Smart Textile Management System, providing a clear and organized view of textile products. It allows users to browse products through proper listing and categorization, making it easy to find required items. Each product includes detailed information such as fabric type, quality, and price, helping users make informed decisions.

The module also supports search and filter options, enabling users to quickly locate products based on their needs. Additionally, it displays product availability status, ensuring users are aware of stock levels in real time. With a user-friendly interface and structured presentation, this module enhances accessibility, improves user experience, and supports efficient product management within the system.

3.4 OUTPUT DESIGN

Output Design refers to the information and results generated by the TexMart – Smart Textile Management System for end users. It is the primary purpose of the system, as users evaluate its effectiveness based on the quality and usefulness of its outputs. The system’s

objectives are reflected through outputs such as inventory reports, billing details, sales summaries, and product information displays.

Outputs in TexMart are presented in various forms, including on-screen displays, reports, and graphical representations for better understanding. They may vary in content, frequency, timing, and format depending on user requirements. Well-designed outputs ensure clarity, accuracy, and timely information, helping users make better decisions. Ultimately, the usefulness of the system is determined by how effectively it delivers meaningful and reliable outputs.

Viewing Product Details

Users can easily browse textile products such as fabrics, yarns, and materials. Each product page displays key details such as fabric type, quality, price, and availability. This organized layout helps users quickly understand product information and make the right selection, improving overall usability.

Stock and Inventory Monitoring

Users can view real-time stock details within the system. The module displays stock levels, product availability, and updates, helping administrators track inventory efficiently. Low stock alerts and stock history provide better control over inventory management and prevent shortages.

Sales and Billing Details

The system provides clear billing information, including invoice details, transaction records, and payment status. Users can easily view and manage sales data, ensuring accurate financial tracking and smooth business operations.

Comprehensive Reports and Analytics

The system generates detailed reports on various aspects of textile management. Sales Reports highlight revenue, product performance, and transaction history. These reports support better decision-making, improve efficiency, and enhance overall business performance in TexMart.

CHAPTER 4

SYSTEM TESTING & IMPLEMENTATION

4.1 SYSTEM TESTING

Testing is an integral part of any system development life cycle. Insufficient and untested applications may lead to system failure, data loss, and reduced performance. Software testing helps in identifying errors, validating system functionalities, and ensuring that the application works according to user requirements.

The first step in system testing is to develop a test plan that tests all aspects of the system. Completeness, correctness, reliability, and maintainability of the software are to be tested for the quality assurance that the system meets the specification and requirements for its intended use and performance. System testing is the most useful practical process of executing a program with the objective of finding errors that make the program fail.

System testing in this project is done in three phases:

- Unit Testing
- Integration Testing
- Validation Testing
- Functional Testing

UNIT TESTING

Unit testing is performed by developers during the development of the TexMart – Smart Textile Management System to identify and fix errors in each module. It is carried out alongside the coding process to ensure that individual modules such as Admin Panel, Supplier and Purchase, Stock Management, and Product Display work correctly. Each module is tested for its functionality to verify that it performs as expected under different conditions.

Forms related to product entry, stock updates, supplier details, and billing are tested to ensure proper data flow and correct execution. If any error occurs, appropriate error messages are

displayed to guide the user. Once all modules function correctly without issues, the system proceeds to the next stage of testing.

INTEGRATION TESTING

Integration testing in the TexMart – Smart Textile Management System is carried out to identify errors in the interaction between different modules. All modules such as Admin Panel, Supplier and Purchase, Stock Management, and Product Display are integrated and tested together as a complete system. During this phase, the data flow and communication between modules are verified to ensure smooth and accurate operation.

Each module is combined and tested to check whether functions like inventory updates, supplier transactions, and product display work correctly when interconnected. The system is thoroughly tested to ensure all components operate as expected without conflicts. The system also undergoes acceptance testing, where it is validated against user requirements. The results confirm that TexMart performs efficiently and meets all specified requirements.

VALIDATION TESTING

Verification and validation testing are essential steps in the development of the TexMart – Smart Textile Management System, performed before the system is delivered to users. These tests ensure that the system meets business requirements and functions as intended. Verification focuses on checking whether each module is developed correctly, while validation ensures the system fulfills its real-world purpose in textile management. Both processes are carried out early to identify and correct errors.

In TexMart, validation testing includes checking various scenarios such as entering invalid or empty data in forms like admin login, product entry, stock updates, and supplier details. For instance, if required fields such as product name or quantity are left empty, the system displays appropriate validation messages to alert the user. This ensures that the system behaves correctly under different conditions and maintains data accuracy and reliability.

FUNCTIONAL TESTING

Functional testing for the TexMart – Smart Textile Management System involves verifying that all core functionalities of the system, such as inventory management, supplier handling, stock updates, and product display, are working as expected. The aim is to ensure that all operations run smoothly and meet user requirements.

Key Areas Tested:

• User Login and Admin Access:Testing secure login, authentication, and role-based access control.

•Product Management: Ensuring products can be added, updated, and deleted correctly in the system.

• Supplier and Purchase Management:Verifying supplier registration, purchase order creation, and transaction records.

• Stock Management: Confirming real-time stock updates, monitoring, and low stock alerts function properly.

• Product Display and Search: Ensuring products are listed correctly with details, and search/filter features work efficiently.

Test cases are designed to validate each function, ensuring the system meets all requirements. The goal of functional testing is to onfirm that TexMart performs all operations accurately and provides a smooth and efficient user experience.

SYSTEM IMPLEMENTATION

The implementation phase of the TexMart – Smart Textile Management System involves deploying the developed system into the working environment and making it ready for use. After the design and development stages, the system was reviewed to ensure it meets the requirements of textile business operations.

Steps in System Implementation:

• Deployment and Integration: The system was deployed in the working environment with smooth integration into existing processes. The development and support teams ensured that all modules functioned properly without disrupting operations.

• Parallel Testing: The system was initially run alongside existing manual methods to verify accuracy and compare results using real-time data. This helped ensure the system operated without errors.

• User Training and Support: Users were trained to handle modules such as Admin Panel, Supplier and Purchase, Stock Management, and Product Display, ensuring easy adoption of the system.

• Post-Implementation Support: Continuous support was provided to resolve minor issues, improve performance, and assist users during initial usage.

The system proved to be user-friendly, efficient, and easy to integrate. User feedback was positive, confirming that TexMart effectively reduces manual work, improves accuracy, and meets the objectives of modern textile management.

CHAPTER 5

CONCLUSION AND FUTURE ENHANCEMENT

CONCLUSION

The TexMart – Smart Textile Management System serves as a modern and efficient platform that transforms the way textile businesses manage their operations. With features such as real-time inventory tracking, supplier and purchase management, stock monitoring, and centralized control through the admin panel, the system creates a seamless integration of technology into textile management. It enhances operational efficiency while ensuring accurate and organized data handling.

This system proves to be an effective solution for textile businesses by offering automation, improved accuracy, and better decision-making support. The user-friendly interface, reliable performance, and well-structured modules ensure ease of use and smooth functionality. TexMart goes beyond traditional management methods, providing a smart digital solution that modernizes textile operations and improves overall productivity.

5.2 FUTURE ENHANCEMENTS

The system has been designed with flexibility and scalability in mind, allowing future improvements and additional features with minimal integration effort. The following enhancements can further improve its functionality:

• Integration of cloud backup for secure data storage and access

• SMS/email notifications for stock updates and purchase activities

• AI-based analytics for better demand prediction and decision-making

• Mobile app development for easy access and real-time monitoring

APPENDICES

APPENDIX 1 -SAMPLE CODE

```
const API_URL = '/api/products';

let products = [];

let cart = []

document.addEventListener('DOMContentLoaded', () => {

    const savedTheme = localStorage.getItem('sreeTheme') || 'light';

    document.documentElement.setAttribute('data-theme', savedTheme)

    const savedCart = localStorage.getItem('sreeCart');

    if (savedCart) { cart = JSON.parse(savedCart); updateCartIcon(); }

    initParticles();

})

function toggleTheme() {

    const current = document.documentElement.getAttribute('data-theme');

    const target = current === 'light' ? 'dark' : 'light';

    document.documentElement.setAttribute('data-theme', target);

    localStorage.setItem('sreeTheme', target);

    showToast(`Switched to ${target} mode`);

}function initParticles() {

    const canvas = document.getElementById('particleCanvas');

    if (!canvas) return;

    const ctx = canvas.getContext('2d');

    canvas.width = window.innerWidth;

    canvas.height = canvas.parentElement.offsetHeight;
```

```
let particlesArray = [];
const numberOfParticles = 60;
class Particle {
    constructor() {
        this.x = Math.random() * canvas.width;
        this.y = Math.random() * canvas.height;
        this.size = Math.random() * 3 + 1;
        this.speedX = Math.random() * 1 - 0.5;
        this.speedY = Math.random() * 1 - 0.5;
    }
    update() {
        this.x += this.speedX;
        this.y += this.speedY;
        if (this.size > 0.2) this.size -= 0.005;
        if (this.x < 0 || this.x > canvas.width) this.speedX = -this.speedX;
        if (this.y < 0 || this.y > canvas.height) this.speedY = -this.speedY;
    }
    draw() {
        ctx.fillStyle = 'rgba(255, 159, 28, 0.4)';
        ctx.beginPath();
        ctx.arc(this.x, this.y, this.size, 0, Math.PI * 2);
        ctx.fill()
function init() {
    for (let i = 0; i < numberOfParticles; i++) {
```

```
            particlesArray.push(new Particle());
        }
    }
    function animate() {
        ctx.clearRect(0, 0, canvas.width, canvas.height);
        for (let i = 0; i < particlesArray.length; i++) {
            particlesArray[i].update();
            particlesArray[i].draw();
        }
        requestAnimationFrame(animate);
    }
    init(); animate()
    window.addEventListener('resize', () => {
        canvas.width = window.innerWidth;
        canvas.height = canvas.parentElement.offsetHeight;
    });
// Intersection Observer for scroll animations
function initScrollReveal() {
    const elements = document.querySelectorAll('.reveal');
    const observer = new IntersectionObserver((entries) => {
        entries.forEach(entry => {
            if (entry.isIntersecting) {
                entry.target.classList.add('active');
                observer.unobserve(entry.target);
```

```
            }
        });
    }, { threshold: 0.1 });
    elements.forEach(el => observer.observe(el));
async function fetchProducts(isShop = true, isIndex = false) {
    try {
        const res = await fetch(API_URL);
        products = await res.json()
        if (isIndex) {
            renderCarousel(products);
        } else {
            populateCategories();
            if (isShop) { renderShop(products); }
            else { renderInventory(products); updateDashboardStats(); }
        }
    } catch (error) {
        showToast('Failed to load products');
    }
}
function getStockBadge(stock) {
    if (stock > 20) return `<span style="font-size:0.85rem; color:var(--primary-light); font-
weight:700;">● In Stock</span>`;
    else if (stock > 0) return `<span style="font-size:0.85rem; color:#fca311; font-
weight:700;">● Low Stock (${stock})</span>`;
```

```
    return `<span style="font-size:0.85rem; color:#ff4d6d; font-weight:700;">● Out of
Stock</span>`;
function getRatingHTML() {
    const stars = Math.floor(Math.random() * 2) + 4; // 4 or 5
    const reviews = Math.floor(Math.random() * 300) + 20;
    let html = '<div class="rating">';
    for (let i = 0; i < 5; i++) { html += i < stars ? '★' : '☆'; }
    html += `<span>(${reviews})</span></div>`;
    return html;
// Render Featured Carousel (Index page)
function renderCarousel(data) {
    const container = document.getElementById('featured-carousel');
    if (!container) return;
    const features = data.slice(0, 4);
    container.innerHTML = '';
    features.forEach((p, idx) => {
        const wrapper = document.createElement('div');
        wrapper.className = 'carousel-card tilt-card reveal';
        wrapper.style.transitionDelay = `${idx * 0.1}s`;
        wrapper.innerHTML = `
      <div class="product-card" onclick="window.location.href='shop.html'">
        <div class="ribbon-wrapper"><div class="ribbon">Top Seller</div></div>
        <div class="img-container">
          <img src="${p.imageUrl}" alt="${p.name}" class="product-image">
```

```
        </div>
        <div class="product-info">
          ${getRatingHTML()}
          <h3 class="product-title">${p.name}</h3>
          <p class="product-price" style="font-size: 1.4rem;">$${p.price.toFixed(2)}</p>
        </div>
      </div>
        container.appendChild(wrapper);
    });
    initScrollReveal();
function renderShop(data) {
    const container = document.getElementById('product-list');
    if (!container) return;
    container.innerHTML = '';
    if (data.length === 0) {
        container.innerHTML = '<p class="reveal" style="grid-column: 1/-1; text-align: center;
color: var(--text-light); font-size: 1.2rem; padding: 3rem;">No products found.</p>';
        initScrollReveal(); return;
    }
    data.forEach((p, idx) => {
        const wrapper = document.createElement('div');
        wrapper.className = 'tilt-card reveal';
        wrapper.style.transitionDelay = `${(idx % 4) * 0.1}s`;
        const ribbonHTML = (p.stock > 100) ? `<div class="ribbon-wrapper"><div
class="ribbon">Bestseller</div></div>` : (p.price > 100) ? `<div class="ribbon-
```

```
wrapper"><div class="ribbon" style="background:var(--
primary);color:#fff;">Premium</div></div>` : "

      wrapper.innerHTML = `

    <div class="product-card" onclick="openQuickView('${p._id}')">

      ${ribbonHTML}

      <div class="img-container">

        <img src="${p.imageUrl}" alt="${p.name}" class="product-image">

        <div class="img-overlay-btn">🔍 Quick View</div>

      </div>

      <div class="product-info">

        <span class="category-tag">${p.category}</span>

        <h3 class="product-title">${p.name}</h3>

        ${getRatingHTML()}

        <p class="product-desc">${p.description.substring(0, 75)}...</p>

        <div class="product-footer">

          <div>

            <div class="product-price">$${p.price.toFixed(2)}</div>

            ${getStockBadge(p.stock)}

          </div>

          <button class="add-to-cart-btn" onclick="event.stopPropagation();
addToCart('${p._id}')" ${p.stock <= 0 ? 'disabled style="background:#ccc;cursor:not-
allowed;"' : ''}>

            <svg width="24" height="24" viewBox="0 0 24 24" fill="none"
stroke="currentColor" stroke-width="2"><line x1="12" y1="5" x2="12"
y2="19"></line><line x1="5" y1="12" x2="19" y2="12"></line></svg>

          </button>
```

```
        </div>
      </div>
    </div>
      container.appendChild(wrapper);
  });
  initScrollReveal();
// Render Inventory Table
function renderInventory(data) {
  const tbody = document.getElementById('inventory-list');
  if (!tbody) return;
  tbody.innerHTML = '';
  if (data.length === 0) {
    tbody.innerHTML = '<tr class="reveal"><td colspan="5" style="text-align: center; padding:2rem;">No items match criteria.</td></tr>';
    initScrollReveal(); return;
  }
  data.forEach((p, idx) => {
    const tr = document.createElement('tr');
    tr.className = 'reveal';
    tr.style.transitionDelay = `${(idx % 10) * 0.05}s`;
    tr.innerHTML = `
    <td><img src="${p.imageUrl}" alt="${p.name}" style="width: 65px; height: 65px; object-fit: cover; border-radius: 10px; box-shadow:var(--shadow-sm);"></td>
    <td>
      <strong style="font-size:1.1rem; color:var(--dark);">${p.name}</strong><br>
```

```
      <span class="category-tag" style="margin-top:0.4rem; font-
size:0.7rem;">${p.category}</span>
    </td>
    <td style="font-weight:900; color:var(--primary); font-
size:1.1rem;">$${p.price.toFixed(2)}</td>
    <td>${getStockBadge(p.stock)}</td>
    <td>
      <div class="action-btns">
        <button class="btn-icon" onclick="editProduct('${p._id}')" title="Edit">✏️</button>
        <button class="btn-icon delete" onclick="deleteProduct('${p._id}')"
title="Delete">🗑️</button>
      </div>
    </td>
      tbody.appendChild(tr);
  });
  initScrollReveal();
}
function openQuickView(id) {
  const p = products.find(x => x._id === id);
  if (!p) return;
  const modal = document.getElementById('quickViewModal');
  const details = document.getElementById('qvDetails');
  details.innerHTML = `
  <h2 style="font-size: 2.5rem; font-weight:900; margin-bottom:0.5rem; color:var(--dark);
line-height:1.2;">${p.name}</h2>
```

```
    <span class="category-tag" style="margin-bottom:1rem;">${p.category}</span>

    ${getRatingHTML()}

    <p style="font-size: 1.1rem; color: var(--text-light); margin-bottom: 2rem; line-height: 1.8;">${p.description}</p>

    <div style="font-size: 3rem; font-weight:900; color:var(--primary); margin-bottom:0.5rem;">$${p.price.toFixed(2)}</div>

    <div style="margin-bottom:2.5rem;">${getStockBadge(p.stock)}</div>

    <button class="btn-primary" style="width:100%; padding:1.2rem; font-size:1.2rem; letter-spacing:1px;" onclick="addToCart('${p._id}'); closeQuickView();" ${p.stock <= 0 ? 'disabled' : ''}>

      <svg width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" style="vertical-align:bottom; margin-right:8px;"><path d="M3 3h2l.4 2M7 13h10l4-8H5.4M7 13L5.4 5M7 13l-2.293 2.293c-.63.63-.184 1.707.707 1.707H17m0 0a2 2 0 100 4 2 2 0 000-4zm-8 2a2 2 0 11-4 0 2 2 0 014 0z"/></svg>

      Add to Shopping Cart

    </button

    document.getElementById('qvImage').src = p.imageUrl;

    modal.classList.add('active');

}

function closeQuickView() {
document.getElementById('quickViewModal').classList.remove('active'); }

function populateCategories() {

    const select = document.getElementById('categoryFilter');

    if (!select) return;

    const categories = [...new Set(products.map(p => p.category))];

    select.innerHTML = '<option value="All">All Categories</option>';

    categories.forEach(c => select.innerHTML += `<option value="${c}">${c}</option>`);
```

```
}
function handleSearch() {
    const term = document.getElementById('searchInput') ?
document.getElementById('searchInput').value.toLowerCase() : '';
    const cat = document.getElementById('categoryFilter') ?
document.getElementById('categoryFilter').value : 'All';
    const sort = document.getElementById('sortFilter') ?
document.getElementById('sortFilter').value : 'default';
    let filtered = products.filter(p => { return (p.name.toLowerCase().includes(term) ||
p.description.toLowerCase().includes(term)) && (cat === 'All' || p.category === cat); });
    if (sort === 'priceLow') { filtered.sort((a, b) => a.price - b.price); }
    else if (sort === 'priceHigh') { filtered.sort((a, b) => b.price - a.price); }
    else if (sort === 'nameAZ') { filtered.sort((a, b) => a.name.localeCompare(b.name)); }
    renderShop(filtered);
}
function handleInventorySearch() {
    const term = document.getElementById('inventorySearch').value.toLowerCase();
    const filtered = products.filter(p => p.name.toLowerCase().includes(term) ||
p.category.toLowerCase().includes(term));
    renderInventory(filtered);
}
function updateDashboardStats() {
    if (!document.getElementById('statsGrid')) return;
    document.getElementById('statTotalProducts').innerText = products.length;
    document.getElementById('statLowStock').innerText = `${products.filter(p => p.stock > 0
&& p.stock <= 10).length} Low / ${products.filter(p => p.stock === 0).length} Out`;
```

```
    document.getElementById('statTotalValue').innerText = `$${products.reduce((sum, p) =>
sum + (p.price * p.stock), 0).toLocaleString(undefined, { minimumFractionDigits: 2 })}`;
}
function toggleCart() {
    document.getElementById('cartSidebar').classList.toggle('active');
    document.getElementById('cartOverlay').classList.toggle('active');
    renderCart();
}
function addToCart(id) {
    const p = products.find(x => x._id === id);
    if (!p || p.stock <= 0) return;
    const existing = cart.find(item => item.product._id === id);
    if (existing) {
        if (existing.quantity < p.stock) { existing.quantity++; showToast('Quantity increased'); }
        else { showToast('Not enough physical stock!', true); }
    } else { cart.push({ product: p, quantity: 1 }); showToast('Added to cart successfully!'); }
    saveCart(); updateCartIcon();
    if (document.getElementById('cartSidebar').classList.contains('active')) renderCart();
}
function updateCartIcon() {
    const badge = document.getElementById('cartCount');
    if (badge) badge.innerText = cart.reduce((sum, item) => sum + item.quantity, 0);
}
function saveCart() { localStorage.setItem('sreeCart', JSON.stringify(cart)); }
function changeQuantity(id, delta) {
```

```
    const item = cart.find(x => x.product._id === id);
    if (item) {
        if (delta > 0 && item.quantity >= item.product.stock) { showToast('Max stock reached', true); return; }
        item.quantity += delta;
        if (item.quantity <= 0) cart = cart.filter(x => x.product._id !== id);
        saveCart(); updateCartIcon(); renderCart();
    }
}
function renderCart() {
    const container = document.getElementById('cartItems');
    const totalEl = document.getElementById('cartTotal');
    if (!container) return;
    container.innerHTML = '';
    if (cart.length === 0) {
        container.innerHTML = '<p class="reveal active" style="text-align:center; margin-top:2rem; font-size:1.2rem; color:var(--text-light);">Your cart is feeling a bit empty!</p>';
        totalEl.innerText = '0.00'; return;
    }
    let total = 0;
    cart.forEach((item, idx) => {
        total += item.product.price * item.quantity;
        container.innerHTML += `
      <div class="cart-item reveal active" style="transition-delay:${idx * 0.05}s">
        <img src="${item.product.imageUrl}" alt="img">
```

```
        <div class="cart-item-info">
          <div class="cart-item-title">${item.product.name}</div>
          <div class="cart-item-price">$${(item.product.price *
item.quantity).toFixed(2)}</div>
          <div class="quantity-ctrl">
            <button class="quantity-btn" onclick="changeQuantity('${item.product._id}', -1)">-
</button>
            <span>${item.quantity}</span>
            <button class="quantity-btn" onclick="changeQuantity('${item.product._id}',
1)">+</button>
          </div>
        </div>
      </div>
    });
    totalEl.innerText = total.toLocaleString(undefined, { minimumFractionDigits: 2 });
}
function checkout() {
    if (cart.length === 0) { showToast('Cart is empty', true); return; }
    const itemsContainer = document.getElementById('cartMainContent');
    const loader = document.getElementById('checkoutLoader');
    const success = document.getElementById('checkoutSuccess');
    itemsContainer.style.display = 'none';
    loader.classList.add('active');
    setTimeout(() => {
        loader.classList.remove('active');
```

```
        success.classList.add('active');
        cart = []; saveCart(); updateCartIcon()
        setTimeout(() => {
            toggleCart();
                setTimeout(() => {
                itemsContainer.style.display = 'block';
                success.classList.remove('active');
                renderCart();
            }, 500);
        }, 2500);
    }, 2000);
}
function openModal(isEdit = false) {
    document.getElementById('productModal').classList.add('active');
    const title = document.getElementById('modalTitle');
    if (!isEdit) {
        title.innerText = 'Add New Product'; document.getElementById('productForm').reset();
document.getElementById('productId').value = '';
    } else { title.innerText = 'Edit Product'; }
}
function closeModal() {
document.getElementById('productModal').classList.remove('active'); }
async function saveProduct(e) {
    e.preventDefault();
    const id = document.getElementById('productId').value;
```

```
    const productData = {
        name: document.getElementById('name').value, category:
document.getElementById('category').value,
        description: document.getElementById('description').value, price:
parseFloat(document.getElementById('price').value),
        stock: parseInt(document.getElementById('stock').value), imageUrl:
document.getElementById('imageUrl').value || 'https://images.unsplash.com/photo-
1620799140408-edc6dcb6d633?auto=format&fit=crop&w=300&q=80'
    };
    const method = id ? 'PUT' : 'POST';
    const url = id ? `${API_URL}/${id}` : API_URL;
    try {
        const res = await fetch(url, { method, headers: { 'Content-Type': 'application/json' },
body: JSON.stringify(productData) });
        if (res.ok) { closeModal(); showToast(id ? 'Product updated!' : 'Product added!');
fetchProducts(false); }
        else { const err = await res.json(); showToast('Error: ' + err.message, true); }
    } catch (error) { }
}
function editProduct(id) {
    const p = products.find(prod => prod._id === id);
    if (p) {
        document.getElementById('productId').value = p._id;
document.getElementById('name').value = p.name;
        document.getElementById('category').value = p.category;
document.getElementById('description').value = p.description;
```

```
        document.getElementById('price').value = p.price;
document.getElementById('stock').value = p.stock;
        document.getElementById('imageUrl').value = p.imageUrl; openModal(true);
    }
}
async function deleteProduct(id) {
    if (confirm('Delete this product permanently?')) {
        try { const res = await fetch(`${API_URL}/${id}`, { method: 'DELETE' }); if (res.ok) {
showToast('Product deleted'); fetchProducts(false); } } catch (error) { }
    }
}
function showToast(message, isError = false) {
    let container = document.getElementById('toastWrapper');
    if (!container) {
        container = document.createElement('div'); container.id = 'toastWrapper';
container.className = 'toast-container'; document.body.appendChild(container);
    }
    const t = document.createElement('div');
    t.className = 'toast';
    if (isError) t.style.borderLeftColor = '#ff4d6d';
    const icon = isError ?
        `<svg width="24" height="24" viewBox="0 0 24 24" fill="none" stroke="#ff4d6d"
stroke-width="2"><circle cx="12" cy="12" r="10"></circle><line x1="15" y1="9" x2="9"
y2="15"></line><line x1="9" y1="9" x2="15" y2="15"></line></svg>` :
```

```
        `<svg width="24" height="24" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" style="color:var(--primary);"><path d="M22 11.08V12a10 10 0 1 1-5.93-9.14"></path><polyline points="22 4 12 14.01 9 11.01"></polyline></svg>`;
    t.innerHTML = `${icon} ${message}`;
    const bar = document.createElement('div');
    bar.style.cssText = `position:absolute; bottom:0; left:0; height:4px; width:100%; background:${isError ? '#ff4d6d' : 'var(--primary)'}; animation:shrink 3s linear forwards;`;
    t.appendChild(bar)
    container.appendChild(t);
    setTimeout(() => {
        t.style.animation = 'slideOutRight 0.5s cubic-bezier(0.165, 0.84, 0.44, 1) forwards';
        setTimeout(() => t.remove(), 500);
    }, 3000);
}
function initGlobalFeatures() {
    const newsletterBtns = document.querySelectorAll('.newsletter-form button');
    newsletterBtns.forEach(btn => {
        btn.addEventListener('click', (e) => {
            e.preventDefault();
            const input = e.target.previousElementSibling;
            if (!input) return;
            if (input.value.includes('@')) {
                showToast('Subscribed to VIP Newsletter successfully!');
                input.value = '';
            } else {
```

```
            showToast('Please enter a valid email address.', true);
        }
    const currentFileName = window.location.pathname.split('/').pop() || 'index.html';
    document.querySelectorAll('.nav-links a').forEach(link => {
        if (link.getAttribute('href') === currentFileName) {
            link.classList.add('active');
        } else {
            link.classList.remove('active');
        }
    });
    const scrollTopBtn = document.createElement('button');
    scrollTopBtn.innerHTML = '↑';
    scrollTopBtn.title = 'Scroll To Top';
    scrollTopBtn.style.cssText = 'position:fixed; bottom:30px; left:30px; width:50px;
height:50px; border-radius:50%; background:var(--primary); color:#fff; border:none;
cursor:pointer; font-size:1.8rem; font-weight:900; box-shadow:var(--shadow-lg); opacity:0;
transition:all 0.4s cubic-bezier(0.165, 0.84, 0.44, 1); z-index:2000; pointer-events:none;
display:flex; justify-content:center; align-items:center; transform:translateY(20px); border:
2px solid var(--bg-color);';
    scrollTopBtn.addEventListener('mouseover', () => { scrollTopBtn.style.transform =
'translateY(-5px) scale(1.1)'; });
    scrollTopBtn.addEventListener('mouseout', () => { scrollTopBtn.style.transform =
'translateY(0) scale(1)'; });
    document.body.appendChild(scrollTopBtn);
indow.addEventListener('scroll', () => {
        if (window.scrollY > 400) {
```

```
            scrollTopBtn.style.opacity = '1';
            scrollTopBtn.style.pointerEvents = 'all';
            if (!scrollTopBtn.matches(':hover')) scrollTopBtn.style.transform = 'translateY(0)
scale(1)';
        } else {
            scrollTopBtn.style.opacity = '0';
            scrollTopBtn.style.pointerEvents = 'none';
            scrollTopBtn.style.transform = 'translateY(20px) scale(0.9)';
        }
    });
    scrollTopBtn.addEventListener('click', () => { window.scrollTo({ top: 0, behavior:
'smooth' }); });
document.addEventListener('DOMContentLoaded', () => { initGlobalFeatures(); });
```

APPENDIX 2 -SAMPLE SCREEN SHOTS

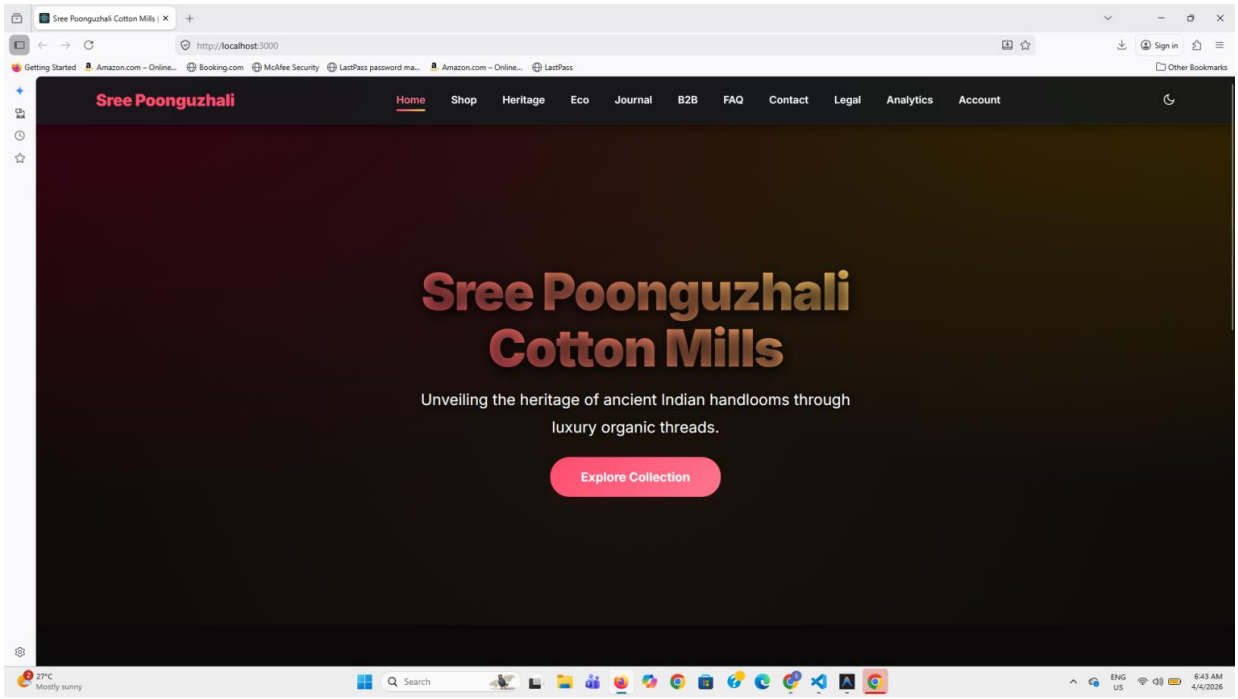


Figure A 2.1 Home Page

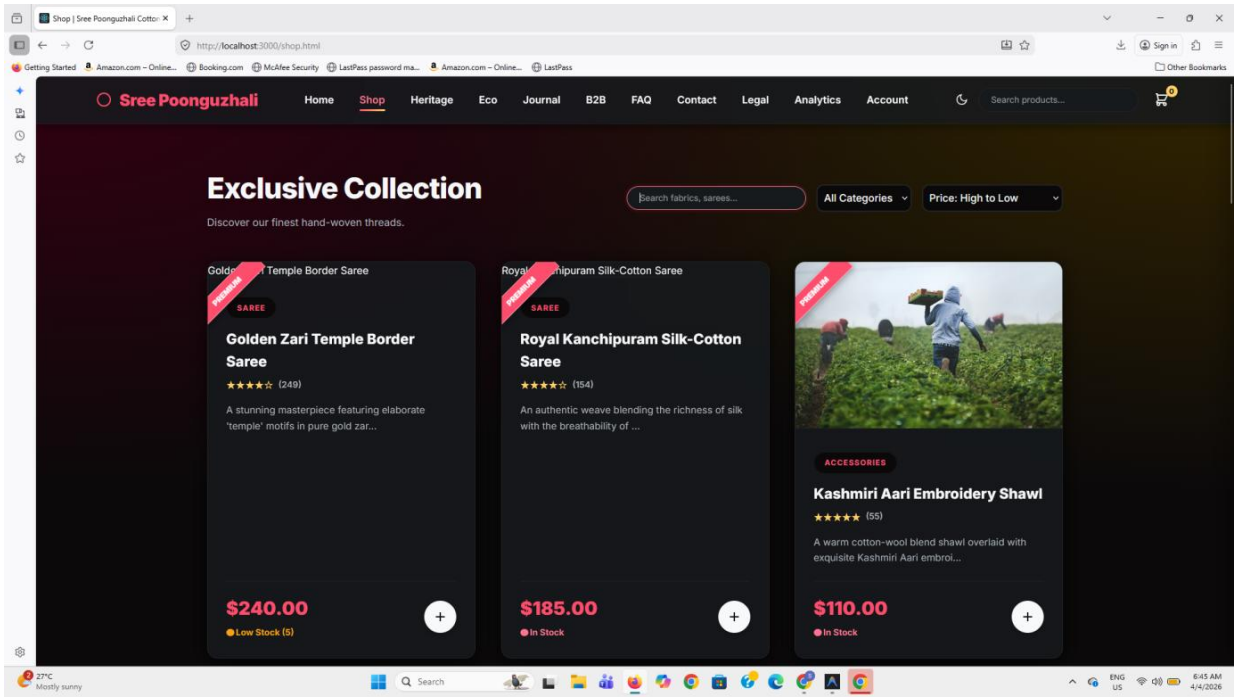


Figure A 2.2 Shop Checklist

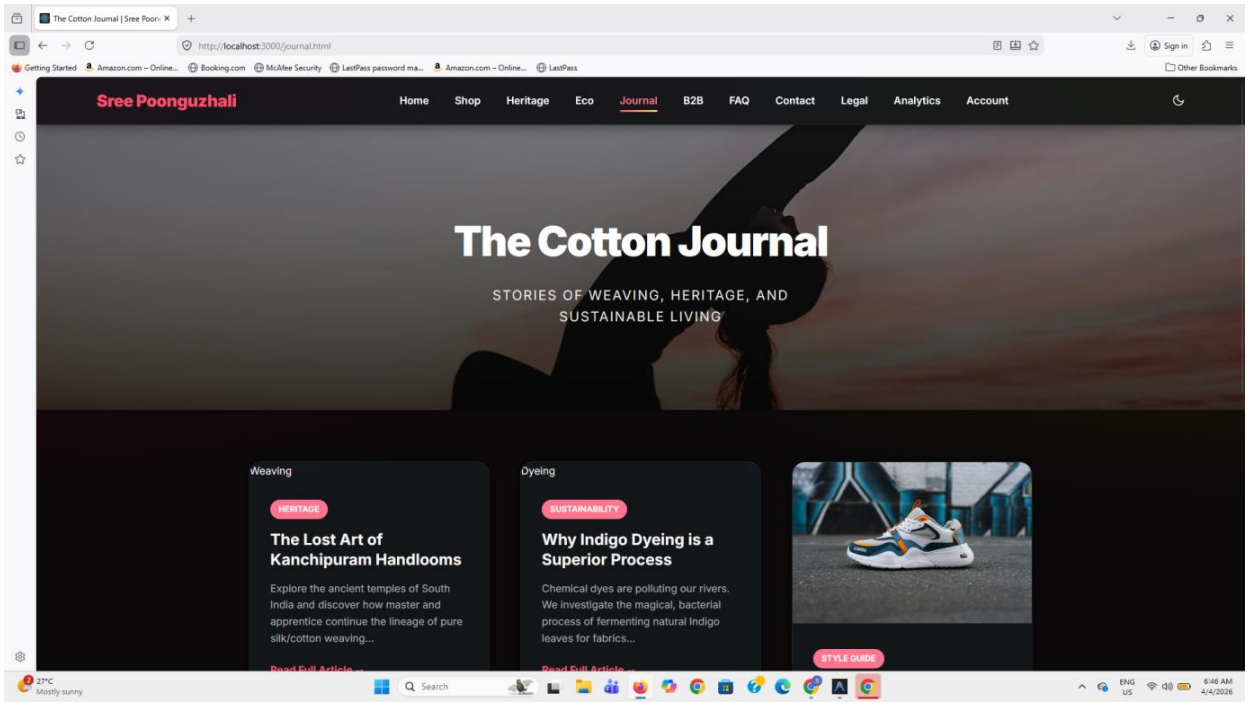


Figure A 2.3 Journal Page

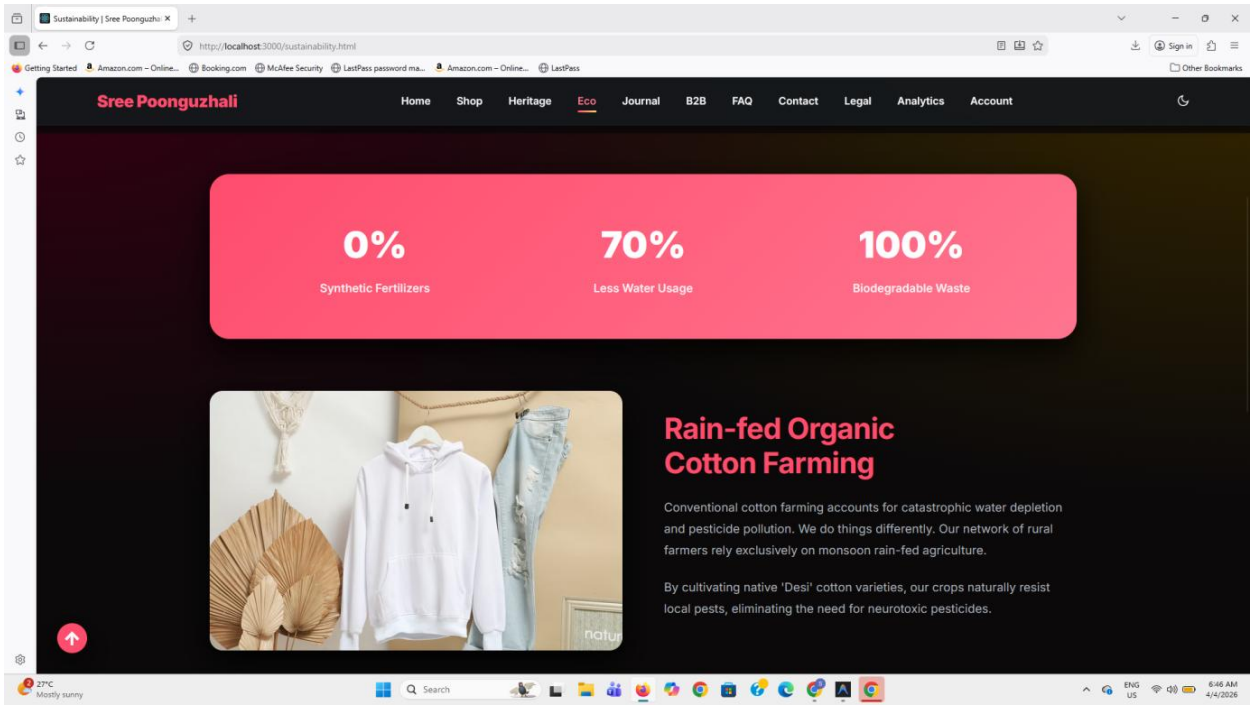


Figure A 2.4 Eco View

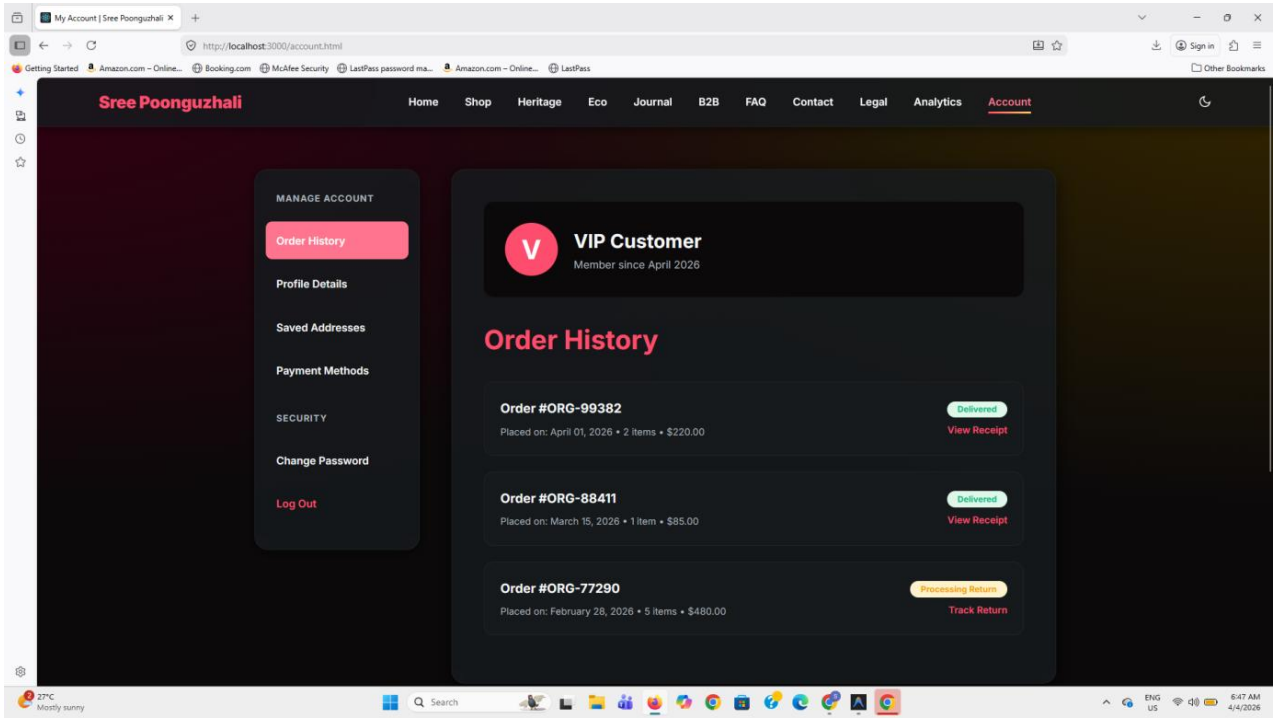


Figure A 2.5 Accounts Page

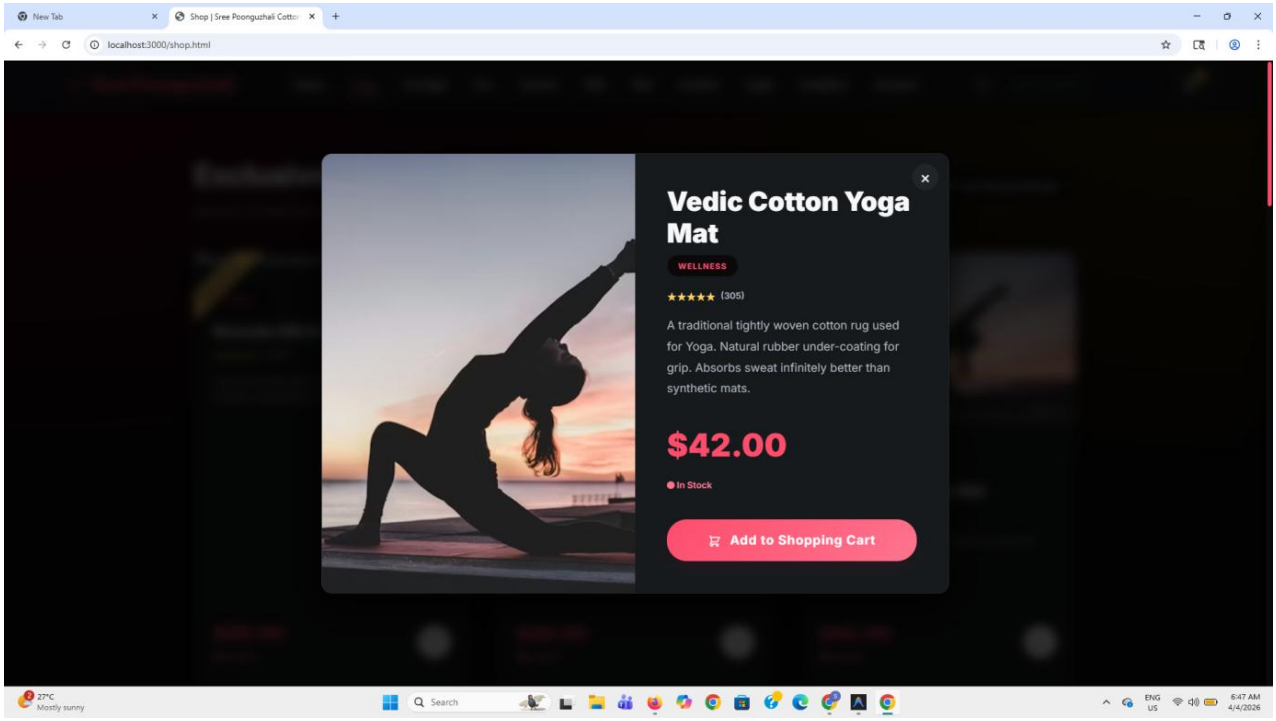


Figure A 2.6 Product Available View

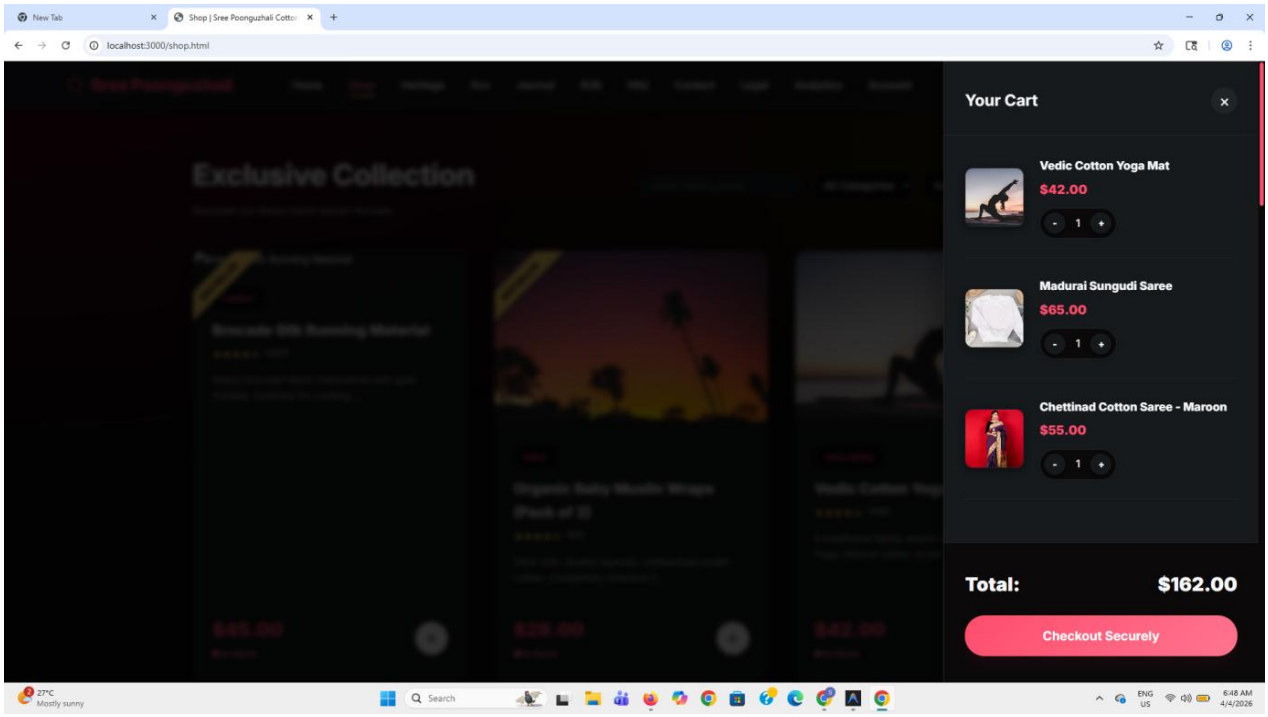


Figure A 2.7 Billing Page

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MAPPING OF PROJECT TO SDG

SDG 8 – Decent Work and Economic Growth

The proposed **TexMart – Smart Textile Management System** strongly supports SDG 8: Decent Work and Economic Growth. The system enhances operational efficiency in textile management by enabling effective tracking of inventory, orders, production processes, and workforce activities. It helps reduce manual work, minimize errors, and improve overall accountability within the organization. By streamlining textile workflows, optimizing resource utilization, and ensuring timely production and delivery, the system contributes to increased productivity, better working conditions, and sustainable economic growth in the textile sector.

SDG 9 – Industry, Innovation and Infrastructure

The project also aligns with SDG 9: Industry, Innovation and Infrastructure by introducing a centralized digital platform for managing textile operations such as inventory, production, orders, and reporting. It promotes innovation through the use of modern web technologies like HTML, CSS, JavaScript, PHP, and MySQL. The system supports digital transformation in the textile industry by replacing traditional manual processes with an automated, efficient, and scalable infrastructure, improving overall operational performance and reliability.

SDG 16 – Peace, Justice and Strong Institutions

This project also supports SDG 16: Peace, Justice and Strong Institutions by enhancing transparency, accountability, and structured workflow management within the Texmart textile system. Features such as inventory tracking, order status updates, production monitoring, and reporting dashboards help maintain clear records and responsibility tracking across operations. This contributes to stronger organizational processes, improved decision-making, and better governance within the textile industry.